



Symbiosis University of Applied Sciences

A Symbiosis Skills University, Indore

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Year:- 1st (2nd Sem)

Branch:- CSIT (B.Tech)

Organisation Name:- Insta Dot Analytics

Internship Designation:- MERN Stack Intern

Mentor's Name:- Dr. Vaishali Joshi

Week:- 1 & 2

Date:- 17/06/2026

INTERNSHIP OVERVIEW

About the Organization

Insta Dot Analytics is a technology-driven company focused on software development, web technologies, and data-driven solutions. The organization provides practical industry exposure to students through project-based learning and hands-on development activities.

Internship Objective

The primary objective of this internship was to gain practical knowledge of Full Stack Web Development using the MERN Stack (MongoDB, Express.js, React.js, and Node.js). The internship focused on developing industry-relevant skills in frontend development, backend development, database management, API integration, authentication, and deployment practices.

Throughout the internship, various assignments and projects were completed to strengthen understanding of modern web application development. The internship provided exposure to real-world development workflows, version control practices, debugging techniques, and collaborative software engineering methodologies.

Week 1 Report (Day 1 - Day 7)

Targets for the Week

- Set up the MERN Stack development environment.
- Learn modern JavaScript concepts.
- Build backend APIs using Express.js.
- Connect applications with MongoDB.
- Develop React frontend applications.
- Understand full-stack integration and testing.

Day-wise Work Completed

Day 1 – JavaScript Fundamentals & UI Development

- Installed and configured Node.js, npm, and VS Code.
- Practiced ES6+ concepts including arrow functions, destructuring, template literals, and promises.
- Developed a JavaScript data processing task.
- Created a responsive user profile card using HTML and CSS.

Day 2 – Express.js Backend Development

- Created an Express.js server.
- Implemented REST API endpoints.
- Worked with Node.js file system modules.
- Tested APIs using browser and Postman.

Day 3 – MongoDB Integration

- Connected Express application with MongoDB.
- Created Mongoose schemas and models.
- Implemented CRUD operations.
- Added middleware and centralized error handling.

Day 4 – React Frontend Development

- Set up React project using Vite.
- Developed reusable React components.
- Connected frontend with backend APIs.
- Implemented data fetching using hooks.

Day 5 – Production Build & Optimization

- Learned environment configuration using .env files.
- Generated optimized production builds.

- Configured static file serving.
- Improved project structure and organization.

Day 6 – Testing & Validation

- Performed end-to-end testing of the application.
- Validated frontend-backend communication.
- Tested database operations and API responses.
- Fixed bugs identified during testing.

Day 7 – State Management

- Implemented React Context API.
- Learned centralized state management.
- Used useReducer for managing application state.
- Improved code quality and project scalability.

Challenges/Roadblocks

- Faced CORS issues during API integration.
- Encountered MongoDB connection and schema validation errors.
- Had difficulties understanding state management concepts initially.

Resolution

- Configured CORS middleware properly.
- Debugged database connections and validation rules.
- Practiced Context API implementation through project tasks.

Learning Outcomes

Technical Skills

JavaScript (ES6+), Node.js, Express.js, MongoDB, Mongoose, React.js, REST APIs, GitHub, Postman, Context API.

Soft Skills

Problem-solving, debugging, self-learning, documentation, time management, and teamwork.

Plan for Next Week

- Learn advanced React concepts.
- Implement authentication and authorization.
- Build e-commerce features.
- Develop shopping cart and wishlist modules.
- Improve application performance.

WEEK 2 REPORT (Day 8 – Day 15)

Targets for the Week

- Develop advanced MERN Stack applications.
- Learn authentication and authorization.
- Implement e-commerce functionalities.
- Optimize React application performance.
- Build complete full-stack project modules.

Day-wise Work Completed

Day 8 – Kanban Task Management Application

- Developed a Kanban board interface.
- Created task cards and workflow columns.
- Implemented task status management.

Day 9 – Undo/Redo Functionality

- Built state history management.
- Implemented undo and redo operations.
- Improved user interaction and usability.

Day 10 – React Performance Optimization

- Used React.memo for component optimization.
- Implemented useMemo and useCallback hooks.
- Reduced unnecessary component re-rendering.

Day 11 – Authentication System

- Implemented user registration and login.
- Integrated JWT authentication.
- Added password hashing using bcrypt.
- Developed protected routes.

Day 12 – Product Management Module

- Created Product CRUD APIs.
- Developed product forms and validation.
- Connected frontend and backend operations.

Day 13 – Shopping Cart Development

- Built shopping cart functionality.
- Implemented quantity management.

- Added automatic price calculations and checkout features.

Day 14 – Wishlist & Routing

- Developed wishlist functionality.
- Implemented dynamic product routing.
- Added user notifications and feedback messages.

Day 15 – Order Management System

- Created order processing APIs.
- Developed order history functionality.
- Integrated checkout with database operations.
- Tested complete e-commerce workflow.

Challenges/Roadblocks

- Managing global state across multiple components.
- Securing authentication and route protection.
- Synchronizing frontend state with backend data.

Resolution

- Used Context API and useReducer.
- Implemented JWT-based authentication.
- Improved API integration and state update logic.

Learning Outcomes

Technical Skills

JWT Authentication, bcrypt, React Router, Context API, useReducer, Product CRUD, Shopping Cart, Wishlist, Order Management, Performance Optimization.

Soft Skills

Analytical thinking, project planning, debugging, code organization, testing, and independent problem-solving.

Plan for Next Week

- Explore deployment and hosting.
- Learn advanced security practices.
- Improve database optimization techniques.
- Work on scalable real-world projects.

TECHNOLOGIES LEARNED

Frontend Technologies

- HTML5
- CSS3
- JavaScript (ES6+)
- React.js
- React Router
- Context API
- Vite

Backend Technologies

- Node.js
- Express.js
- REST API Development
- Middleware Implementation

Database Technologies

- MongoDB
- Mongoose ODM

Development Tools

- Visual Studio Code
- Git
- GitHub
- Postman
- MongoDB Compass

Key Technical Concepts Learned

- MERN Stack Architecture
- CRUD Operations
- API Integration
- State Management
- Authentication and Authorization
- JWT Token-Based Security
- Password Hashing using bcrypt
- Frontend Optimization Techniques
- Error Handling and Debugging
- Version Control Practices

LEARNING OUTCOMES

Technical Skills Acquired

During the internship, I gained practical experience in developing full-stack web applications using the MERN Stack. I learned how to create responsive user interfaces using React.js, build RESTful APIs using Express.js, manage databases using MongoDB, and connect frontend applications with backend services.

I also developed skills in implementing authentication systems, managing application state, handling API requests, debugging applications, and optimizing application performance.

Professional Skills Acquired

The internship helped improve several professional skills including:

- Problem Solving
- Time Management
- Analytical Thinking
- Technical Documentation
- Team Collaboration
- Self-Learning
- Communication Skills
- Software Development Best Practices

CHALLENGES FACED AND SOLUTIONS

Challenge 1: Understanding MERN Stack Architecture

Initially, understanding the interaction between the frontend, backend, and database components was challenging.

Solution

Through hands-on assignments, API testing, and practical implementation, I developed a clear understanding of the complete request-response cycle within a MERN Stack application.

Challenge 2: API Integration Issues

While connecting the frontend and backend applications, several issues related to API communication and CORS policies were encountered.

Solution

These issues were resolved by properly configuring middleware, validating API endpoints, and testing requests using Postman.

Challenge 3: Database Connectivity and Validation

Errors occurred while connecting MongoDB and implementing schema validations.

Solution

Database schemas were carefully designed using Mongoose, and validation mechanisms were implemented to ensure data consistency.

Challenge 4: State Management in React

Managing data across multiple components became increasingly complex as the application grew.

Solution

React Context API and useReducer were utilized to centralize state management and improve application scalability.

CONCLUSION

The internship at Insta Dot Analytics provided valuable practical exposure to modern web development technologies and industry workflows. Through various assignments and project-based activities, I gained hands-on experience in frontend development, backend API development, database management, authentication systems, and full-stack application architecture.

The internship successfully bridged the gap between academic learning and industry practices by allowing me to apply theoretical concepts in real-world development scenarios. It significantly enhanced my technical knowledge, problem-solving abilities, and confidence in developing complete web applications using the MERN Stack.

Overall, the internship was a highly productive learning experience that contributed to both my technical growth and professional development.

SIGNATURES

Industry Supervisor Signature: _____

University Faculty Guide Signature: _____

Date: _____